STPM Physics · Semester 2 · Chapter 13

ELECTROSTATICS

13.1 Coulomb's Law

DEFINITIONS & LAWS

- **Coulomb's law** states that the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.
- The force acts along the line joining the two charges; it is repulsive for like charges and attractive for unlike charges. The two forces form a Newton's third law pair (equal in magnitude, opposite in direction).
- **Principle of superposition**: the net force on a charge is the vector sum of the forces exerted on it by each of the other charges separately.
- **Relative permittivity** ε_r of a medium is the ratio of the force between two charges in a vacuum to the force between the same charges, at the same separation, in the medium ($\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$).
- **Charge is quantised**: $Q = ne$, where n is an integer.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Coulomb's law (vacuum)	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2} = \frac{kQ_1Q_2}{r^2}$	F = electrostatic force (N); Q_1, Q_2 = point charges (C); r = separation (m); $k = \frac{1}{4\pi\varepsilon_0} = 8.99\times10^9$ N m^2 C^{-2}; $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}
Coulomb's law in a medium	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0\varepsilon_r r^2}$	ε_r = relative permittivity of the medium (no unit); other symbols as above
Force in an electric field	$F = qE$	F = force (N); q = charge (C); E = electric field strength (N C^{-1} or V m^{-1})
Acceleration of a charge	$a = F/m$	a = acceleration (m s^{-2}); F = force (N); m = mass (kg)
Charged spheres on threads	$\tan\theta = \frac{F}{mg}$	Two small identical charged spheres on threads; θ = angle of thread to the vertical; F = horizontal electric force (N), balanced by the tension and the weight mg; m = mass (kg); g = acceleration of free fall (m s^{-2})

- $e = 1.60\times10^{-19}$ C; 1 μC = 10^{-6} C; 1 nC = 10^{-9} C.

TIPS TO ANSWER

- State the law or formula first, then substitute with all quantities in SI units (convert μC, nC, cm, mm), then give the answer with unit to 3 s.f.
- State the nature of the force: "repulsive" if the charges have the same sign, "attractive" if the signs are opposite.
- For several charges, draw the force on the chosen charge from each of the others, add them as vectors (along a line: take one direction as positive) and state the direction of the resultant.
- For a charged-sphere or thread problem, draw the free-body diagram (T, mg, F), resolve and write $\tan\theta = \frac{F}{mg}$.
- For a null point, equate the magnitudes of the two forces (or fields) and solve for the distance.
- Use ε_0 from the list of constants $\frac{1}{4\pi\varepsilon_0}$, not a rounded $k = 9 \times 10^9$ N m^2 C^{-2}. A slightly different final answer from the marking scheme can be marked wrong. For π, use the calculator value or 3.142.
- Do not substitute the signs of the charges into Coulomb's law; use magnitudes and find the direction of the force with a positive test charge (like charges repel, unlike attract).

SECTION A TRAPS

Trap: Missing square: using $F \propto \frac{1}{r}$ instead of $\frac{1}{r^2}$. Doubling r gives $\frac{F}{4}$ (not $\frac{F}{2}$); tripling r gives $\frac{F}{9}$.

Trap: Unconverted prefixes: using 3 μC as 3 or 3×10^{-3} C, or 30 cm as 30 m, gives answers wrong by powers of ten; always check the order of magnitude.

Trap: Wrong factor in a combined change: doubling one charge and halving the separation gives 2 × 4 = 8F (not 4F); doubling one charge and tripling r gives $\frac{2F}{9}$.

Trap: Medium: in a medium ε_r divides the force (F decreases); a distractor multiplies by ε_r instead.

TRICKS

Trick: Ratio method: write $\frac{F'}{F} = \frac{Q_1'Q_2'}{Q_1Q_2}\left(\frac{r}{r'}\right)^2$ and substitute the factors directly; no constants are needed.

Trick: Null point between two like charges: it lies nearer the smaller charge, and $\frac{r_1}{r_2} = \sqrt{\frac{Q_1}{Q_2}}$; for +Q and +4Q the ratio of distances is 1 : 2.

13.2 Electric Field

DEFINITIONS & LAWS

- An **electric field** is a region in which an electric force acts on a charged body.
- **Electric field strength** at a point is the force per unit positive charge placed at that point ($E = \frac{F}{q}$); it is a vector quantity whose direction is that of the force on a positive test charge.
- **Electric field line**: a line whose tangent at any point gives the direction of the electric field at that point; the density of the lines is proportional to the field strength.
- Field lines begin on positive charges (or at infinity) and end on negative charges (or at infinity); they never cross, because the field has only one direction at a point.
- The field of a point charge is radial: outward for a positive charge and inward for a negative charge.
- A **uniform field** exists between two oppositely charged parallel plates: the field lines are parallel and equally spaced, and E has the same value at every point (apart from the edges).
- **Superposition**: the resultant field at a point is the vector sum of the fields due to the individual charges.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Electric field strength	$E = \frac{F}{q}$	E = electric field strength (N C^{-1} = V m^{-1}); F = force on the test charge (N); q = test charge (C)
Field of a point charge	$E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{kQ}{r^2}$	Point charge, or outside a charged sphere; Q = charge (C); r = distance from the centre (m)
Field between parallel plates	$E = \frac{V}{d}$	V = potential difference (V); d = plate separation (m)
Field and potential gradient	$E = -\frac{dV}{dr}$	$\frac{dV}{dr}$ = potential gradient (V m^{-1}); the field points in the direction of decreasing potential
Force and acceleration of a charge	$F = qE;\ a = \frac{qE}{m}$	F = force (N); a = acceleration (m s^{-2}); q = charge (C); m = mass (kg)
Charge starting from rest	$v^2 = 2as$	Charge starting from rest with uniform acceleration; v = final speed (m s^{-1}); s = distance (m)
Charge entering perpendicular to field	Parabolic path	The path is a parabola, with constant velocity component along the plates
Field near a charged conducting plate	$E = \frac{\sigma}{\varepsilon_0}$	Near a large charged conducting plate; σ = surface charge density (C m^{-2}); ε_0 = permittivity of free space (F m^{-1})

TIPS TO ANSWER

- Calculate the field of each charge separately, give each a direction, then add as vectors; end with the magnitude and direction of the resultant.
- For a charge in a field state that the force is F = qE, which is in the direction of E for a positive charge and opposite to E for a negative charge, then use a = F/m.
- Draw field lines with arrows, with correct start/end on charges, no crossing, and closer spacing where the field is stronger.
- State the direction of E clearly: "away from the positive charge" or "towards the negative charge".
- Electric field strength is a vector: find each field by placing a positive test charge at the point, then add the fields as vectors (draw them). Example: midway between −5 μC and +5 μC, both fields point towards the negative charge, so they add.

SECTION A TRAPS

Trap: Field direction: at the midpoint between +Q and −Q the fields add (same direction, towards −Q) so E = $\frac{2kQ}{r^2}$, not zero; for two equal like charges the fields cancel at the midpoint.

Trap: Inverse proportionality: E = $\frac{kQ}{r^2}$ falls as $\frac{1}{r^2}$; a distractor uses $\frac{1}{r}$. Close to a large plate the field is constant, not dependent on r.

Trap: Electron motion: an electron accelerates opposite to the direction of E; a = $\frac{eE}{m_e}$, using m_e = 9.11×10^{-31} kg and not the mass of a proton.

Trap: Field lines: options claiming that lines may cross, or that the tangent gives the velocity of the charge, are incorrect.

TRICKS

Trick: Null-point test: for unlike charges the field is zero only outside the pair, on the side of the smaller charge; for like charges it is zero between them, nearer the smaller charge.

Trick: A zero potential does not mean zero field, and a zero field does not mean zero potential; keep the two quantities separate.

13.3 Gauss's Law

DEFINITIONS & LAWS

- **Electric flux** through a surface is the product of the electric field strength normal to the surface and the area of the surface, $\Phi = EA\cos\theta$, where θ is the angle between E and the normal to the surface.
- **Gauss's law** states that the total electric flux through any closed surface is equal to the net charge enclosed by the surface divided by ε_0.
- The flux depends only on the net enclosed charge, not on the shape or size of the closed surface, and not on the position of the charge inside the surface.
- A charge outside the closed surface contributes zero net flux, because every field line that enters the surface also leaves it.
- In electrostatic equilibrium the field inside the material of a conductor is zero, so the excess charge lies on the surface of the conductor.
- **Surface charge density** σ is the charge per unit area of a surface, $\sigma = \frac{Q}{A}$.
- Outside a charged sphere the field is the same as if all the charge were concentrated at the centre.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Gauss's law	$\Phi = \oint E \cdot dA = \frac{\Sigma Q}{\varepsilon_0}$	Φ = electric flux (N m^2 C^{-1} = V m); E = electric field strength (N C^{-1}); A = area (m^2); ΣQ = algebraic sum of the enclosed charges (C); ε_0 = permittivity of free space (F m^{-1})
Field of a point charge or sphere	$E = \frac{Q}{4\pi\varepsilon_0 r^2}$	Valid for r ≥ radius R; Q = charge (C); r = distance from the centre (m)
Field of an infinite line charge	$E = \frac{\lambda}{2\pi\varepsilon_0 r}$	λ = charge per unit length (C m^{-1}); r = perpendicular distance from the line (m)
Field of an infinite plane sheet	$E = \frac{\sigma}{2\varepsilon_0}$	σ = charge per unit area (C m^{-2})
Field at a conductor surface	$E = \frac{\sigma}{\varepsilon_0}$	Just outside the surface of a conductor; σ = charge per unit area (C m^{-2})

- $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}.

TIPS TO ANSWER

- State Gauss's law, choose a Gaussian surface with the symmetry of the charge distribution (sphere, coaxial cylinder or pillbox) and justify that E is constant over it and perpendicular to it.
- Write Φ = E × (area of surface) = $\frac{Q_{enclosed}}{\varepsilon_0}$, then solve for E and state its dependence ($\frac{1}{r^2}$, $\frac{1}{r}$ or constant).
- For flux questions add the enclosed charges algebraically with their signs before dividing by ε_0.
- For a conductor state: E = 0 inside, because the enclosed charge for a Gaussian surface in the conductor is zero, so the charge resides on the outer surface.

- To show that Gauss's law is equivalent to Coulomb's law: draw a Gaussian sphere round a point charge Q, apply Gauss's law, use the definition of electric flux ($\Phi = E \times 4\pi r^2$) to get E, then place a charge q and use $E = \frac{F}{q}$ to obtain Coulomb's law.

SECTION A TRAPS

Trap: Surface size: doubling the radius of a sphere around a central charge leaves the flux unchanged; options with $\frac{1}{4}$ or $\frac{1}{2}$ are wrong because they confuse E with Φ.

Trap: Charge outside: a charge not enclosed gives zero net flux however near it is; distractors give $\frac{Q}{\varepsilon_0}$ or $\frac{Q}{2\varepsilon_0}$.

Trap: Sign slip: ΣQ must be algebraic; +3Q, −Q and +2Q give $\frac{4Q}{\varepsilon_0}$, not $\frac{6Q}{\varepsilon_0}$.

Trap: Field forms: the line-charge field is proportional to $\frac{1}{r}$ (not $\frac{1}{r^2}$), and the plane sheet field is independent of r.

TRICKS

Trick: Symmetry fractions: a charge at the centre of a cube gives $\frac{Q}{\varepsilon_0}$ through the whole cube, so $\frac{Q}{6\varepsilon_0}$ through each face; a charge at one corner shared by 8 cubes gives $\frac{Q}{8\varepsilon_0}$ through each cube.

Trick: Total flux through a closed surface never contains r: reject any option such as $\frac{Q}{4\pi\varepsilon_0 r^2}$, which is the field E and not the flux.

13.4 Electric Potential

DEFINITIONS & LAWS

- **Electric potential** at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point ($V = \frac{W}{q}$).
- **Potential difference** between two points is the work done per unit positive charge in moving a test charge from one point to the other.
- **Electric potential energy** of a charge at a point is the work done in bringing the charge from infinity to that point ($U = qV$).

- An **equipotential surface** is a surface on which all points are at the same potential; no work is done in moving a charge along it, and it is perpendicular to the electric field lines.
- The electric field is the negative potential gradient ($E = -\frac{dV}{dr}$): the field points in the direction of decreasing potential, and its magnitude is large where equipotentials are close together.
- Potential is a scalar: the potential due to several charges is the algebraic sum of the individual potentials.
- One **electron-volt** (eV) is the energy gained by an electron when it is accelerated through a potential difference of 1 V; 1 eV = 1.60×10^{-19} J.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Potential of a point charge	$V = \frac{Q}{4\pi\varepsilon_0 r} = \frac{kQ}{r}$	V = electric potential (V); Q = charge (C); r = distance from a point charge, or from the centre of a charged sphere for r ≥ R (m)
Work done by an external force	$W = q(V_B - V_A) = q\Delta V$	Work done by an external force in moving charge q from A to B (J); q = charge (C); V_A, V_B = potentials at A and B (V)
Work done by the field	$W = q(V_A - V_B)$	Work done by the field in moving charge q from A to B (J)
Electric potential energy	$U = qV$	U = potential energy (J); q = charge (C); V = potential (V)
Potential energy of two point charges	$U = \frac{kQ_1Q_2}{r}$	Q_1, Q_2 = point charges (C); r = separation (m); $k = \frac{1}{4\pi\varepsilon_0}$
Field and potential	$E = \frac{V}{d}$; $E = -\frac{dV}{dr}$	$E = \frac{V}{d}$ for a uniform field; $E = -\frac{dV}{dr}$ in general; E = field strength (V m^{-1}); d = separation (m)
Charge accelerated through p.d.	$\frac{1}{2}mv^2 = qV$	Charge accelerated from rest through potential difference V; m = mass (kg); v = speed (m s^{-1}); q = charge (C); V = potential difference (V)
Charged oil drop at rest	$\frac{qV}{d} = mg$	Charged oil drop held at rest between plates; q = charge (C); V = p.d. (V); d = plate separation (m); m = mass (kg); g = acceleration of free fall (m s^{-2})

- 1 eV = 1.60×10^{-19} J.

TIPS TO ANSWER

- State the formula, substitute with signs for each charge ($V = \frac{Q}{4\pi\varepsilon_0 r}$ keeps the sign of Q), add algebraically, and give the answer in V with the sign stated.

- For work done, state who does it: by the field $q(V_A - V_B)$, by an external force $q(V_B - V_A)$; give the sign and unit (J).
- For a drop in balance write $qE = mg$ with $E = \frac{V}{d}$, using the total potential difference between the plates and the separation in metres.
- For equipotentials sketch concentric circles (spheres) round a point charge, with field lines radial and perpendicular to them.
- Potential is a scalar, so substitute the signs of the charges and add algebraically. Example: midway between −5 µC and +5 µC, $V = 0$ although E is not zero.
- Two identical charged spheres touched and then separated share the total charge equally. Spheres joined by a wire exchange charge until their potentials are equal.

SECTION A TRAPS

Trap: Vector addition of potentials: V is a scalar; +4 µC and −4 µC at equal distances give $V = 0$ at the midpoint, though $E \neq 0$ there; a distractor gives $\frac{2kQ}{r}$.

Trap: Sign of work: a positive charge moving from high to low potential has positive work done by the field; the external work is of opposite sign.

Trap: Potential difference between plates: plates at +150 V and −150 V have $\Delta V = 300$ V, not 150 V.

Trap: Direction: potential decreases along the field direction; a distractor says it increases or is undefined away from point charges.

TRICKS

Trick: Zero-V trick: equal and opposite charges at equal distances give $V = 0$; check the sign of each term before adding.

Trick: Distance check: for the centre of a square of side d, each corner is at $\frac{d}{\sqrt{2}}$, so the factor $\sqrt{2}$ appears ($V = 4\sqrt{2}\ \frac{kQ}{d}$ for four equal charges).